

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

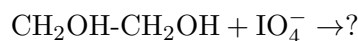
Paper No. CHB-501: Analytical Chemistry-I

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 which is compulsory.***Question 1.** Answer all parts of the following: [5 × 2 = 10]

- (a) What is the difference between specific and selective organic precipitants?
- (b) What is the role of oxine in gravimetric analysis?
- (c) The water of crystallization in $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ could be determined by thermogravimetric analysis. Explain.
- (d) Complete the following Malaprade reaction:



- (e) Brief the key features of Good Laboratory Practices (GLP) and Good Manufacturing Practices (GMP).

Question 2.

- (a) Discuss various steps involved in gravimetric analysis. What is the von Weimarn ratio? Define various terms used in it. [8]
- (b) Write the use of any two of the following organic precipitants in chemical analysis: (i) Dithizone, (ii) Oxine, and (iii) Cupferron. [6]

Question 3.

- (a) What is the isomorphous replacement in the development of precipitates? Explain. [4]
- (b) How can you minimize the magnitude of determinate errors in chemical analysis? [4]
- (c) Differentiate between accuracy and precision. Calculate the absolute and relative precision parameters of the following set of results obtained during the determination of the chromium content of industrial wastewater: 6.8, 7.7, 7.2, 7.3, and 7.5 ppm. [6]

Question 4.

- (a) Discuss the chemistry and stoichiometry of the estimation of vicinal diols using periodic acid. How do you standardize this reagent? [8]

- (b) How can you estimate α -hydroxyamines using Malaprade reagent? [6]
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Question 5. How do you estimate the following? Write reactions: [6 + 4 + 4 = 14]

- (a) Metal ions using potassium bromate as a redox titrant
(b) Nitrite using Chloramine-T
(c) Anhydrides using Karl Fischer reagent